

Forensic Analysis of Cryptocurrency-Based Ransomware

Technical Tracing Methods, Criminal-Justice Challenges,
and a Regional Perspective on the Gulf

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Abstract

Ransomware is one of the most damaging forms of cybercrime, and cryptocurrency is the payment channel that makes it work at scale. This paper reviews how ransomware operations use cryptocurrency for payment and laundering, the technical methods investigators use to trace those funds on public blockchains, and the criminal-justice challenges that follow. Rather than presenting original case statistics, it synthesizes published data and the peer-reviewed tracing literature, and it uses a small number of well-documented public cases, above all the 2021 Colonial Pipeline seizure, to show how the methods work in practice.

The review draws four main conclusions. First, the ransomware payment ecosystem has professionalized around a Ransomware-as-a-Service (RaaS) model with revenue-sharing between developers and affiliates, a structure made visible on-chain by the recovery of the Colonial Pipeline ransom, where blockchain analysis attributed roughly 87 percent of the payment to an affiliate address. Second, most ransomware still relies on Bitcoin, whose public ledger is pseudonymous rather than anonymous, so clustering and flow-tracing remain effective; the shift toward privacy coins such as Monero raises the cost of investigation but pushes it toward the regulated on-ramps and off-ramps where funds enter and leave. Third, attribution strong enough for prosecution almost never comes from the blockchain alone; it comes from correlating on-chain evidence with communications, infrastructure and financial evidence, which is a criminal-justice and cooperation problem as much as a technical one. Fourth, published figures show the ecosystem under real pressure: on-chain ransomware payments were about USD 820 million in 2025, down roughly 8 percent year on year, with only about 28 percent of victims paying, the lowest share on record. The paper adds a regional section on the Gulf, where Saudi and UAE frameworks and cyber-threat-intelligence requirements shape how financial institutions and investigators respond, and it closes with recommendations for investigators, prosecutors, policymakers and the cryptocurrency industry.

Keywords: ransomware; cryptocurrency; blockchain forensics; Bitcoin tracing; Monero; Ransomware-as-a-Service; digital evidence; criminal justice; Gulf Cooperation Council

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1 Introduction

Ransomware encrypts a victim's data, or steals and threatens to publish it, and demands payment for its release. Over the past decade it has grown from opportunistic attacks on individuals into an industry that targets hospitals, pipelines, food producers and government services. Two attacks in 2021 made the scale plain: the DarkSide attack on Colonial Pipeline disrupted fuel supply along the United States East Coast [1], and an attack on JBS disrupted meat processing across several countries. Cryptocurrency is what allows this industry to collect money across borders outside the traditional banking system [3].

Two facts about cryptocurrency shape everything that follows. First, most ransomware payments are made in Bitcoin, and the Bitcoin ledger is public and permanent, so payments are pseudonymous rather than anonymous and can be traced [4, 5]. Second, criminals respond to this transparency by laundering funds through mixers, cross-chain swaps and privacy coins, and by cashing out through exchanges in permissive jurisdictions [6]. Investigation is therefore a moving contest between laundering techniques and tracing methods, played out on top of a legal system that was not designed for borderless digital money.

This paper reviews that contest. It is written for readers who sit across the technical and criminal-justice divide, and it deliberately relies on public, verifiable sources rather than on private case files.

1.1 Objectives

1. To describe how cryptocurrency is used across the ransomware payment lifecycle, from payment to laundering to cash-out.
2. To explain the main technical methods for tracing and attributing ransomware payments on public blockchains, and their limits.
3. To set out the criminal-justice challenges of investigating and prosecuting these crimes, using documented cases.
4. To relate these findings to the regulatory context of Saudi Arabia and the Gulf Cooperation Council (GCC).

1.2 Scope and method

This is a review and synthesis, not a study of original casework. It draws on three kinds of public source: the peer-reviewed literature on cryptocurrency tracing and ransomware economics; annual measurement reports from blockchain-analytics firms, used with attribution and appropriate caution; and official documents such as court filings and press releases for specific cases. It contains no confidential data and analyses no private wallets. Statements about law are drawn from published legislation and legal commentary and are not legal advice. The limitations of this approach are set out in Section 11.

The paper does not provide any operational guidance that would help commit or conceal a crime. It describes investigative methods at the level already established in the public literature, in order to explain how offenders are caught, not how to evade capture.

2 Background and Literature Review

2.1 Ransomware evolution

Ransomware dates to the 1989 AIDS Trojan, distributed on floppy disks, but modern ransomware is defined by two developments: strong asymmetric cryptography that makes decryption without the attacker’s key infeasible, and a criminal division of labour [16]. Contemporary operations separate the roles of malware developer, affiliate who carries out intrusions, initial-access broker who sells footholds, and launderer who moves the proceeds [11]. Target selection has moved from mass campaigns to “big game hunting” against organizations that can pay large sums, and many groups now combine encryption with data theft and the threat of publication, so-called double extortion.

2.2 Ransomware-as-a-Service

The most important structural change is the Ransomware-as-a-Service (RaaS) model, in which a core group provides malware and infrastructure to affiliates in return for a share of each ransom [11]. This lowers the technical barrier to entry and creates an on-chain signature: ransom payments are split between affiliate and developer addresses according to a revenue share. That split is not a hypothesis; it was directly observed in the Colonial Pipeline case (Section 5).

2.3 Cryptocurrency traceability

The foundational result for investigators is that Bitcoin is pseudonymous, not anonymous. Reid and Harrigan showed that the transaction graph can be analysed to link addresses [4], and Meiklejohn and colleagues demonstrated practical clustering: addresses used together as inputs to a transaction are very likely controlled by the same entity, and interactions with known services can attach real-world identities to clusters [5]. These heuristics, co-spend clustering and change-address identification, remain the backbone of blockchain forensics.

Privacy technologies complicate this picture. Mixers and tumblers pool funds from many sources to break the link between input and output, though they can be partially defeated by graph and timing analysis [7]. Privacy coins go further: Monero hides sender, receiver and amount using ring signatures, stealth addresses and confidential transactions [8], and Zcash offers shielded transactions using zero-knowledge proofs [9]. Research on Monero has found that early transactions were sometimes traceable through implementation weaknesses, but the protocol’s design places real limits on chain analysis [8]. Investigators also trace funds across blockchains, since offenders convert between currencies to break the trail, and cross-ledger tracing is an active research area [10].

Table 1: How the main cryptocurrencies used in ransomware differ for investigators. “Traceability” means the practical difficulty of following funds on the public ledger.

Asset	Privacy design	Traceability for investigators	Role in ransomware
Bitcoin	None; public, pseudonymous ledger	High: clustering and flow tracing work well	Dominant payment currency; easiest for victims to obtain
Monero	Ring signatures, stealth addresses, hidden amounts	Very low on-chain; work at entry and exit points instead	Growing use for laundering and, increasingly, ransom demands
Zcash	Optional shielded transactions (zero-knowledge)	Low when shielded, but most use stays transparent	Occasional; shielded use is a minority
Stablecoins	None; public ledger on the host chain	High, and issuers can freeze addresses	Used in laundering and cross-chain hops

2.4 Ransomware economics and measurement

Several studies analyse the economics of ransomware, including how operators price demands and how victims decide whether to pay [12]. For aggregate scale, the most widely cited figures come from blockchain-analytics firms that attribute on-chain flows to known ransomware clusters. These estimates are lower bounds, are revised upward over time as more addresses are attributed, and depend on proprietary attribution, so they should be read as indicative rather than exact. With that caveat, the reported trend is informative and is discussed in Section 8.

2.5 Criminal-justice challenges

The legal literature identifies recurring obstacles: cryptocurrencies are classified inconsistently across jurisdictions [13]; investigations span many countries, and mutual legal assistance is slow [14]; attribution strong enough for court requires more than a chain of addresses; and many agencies lack the tools and trained staff to investigate crypto crime [15]. Successful cases have repeatedly depended on collaboration between law enforcement, exchanges and analytics firms.

2.6 Research gap and this paper’s contribution

The technical and criminal-justice literatures largely run in parallel. This paper connects them for a practitioner audience, grounds the connection in documented public cases rather than invented statistics, and adds a regional analysis for Saudi Arabia and the GCC that the existing literature does not cover.

3 Cryptocurrency in the Ransomware Lifecycle

3.1 Payment

A ransomware operation must collect payment reliably from victims who are, by definition, in crisis. Mature operations use unique payment addresses per victim, automated confirmation that watches the blockchain for the incoming transaction, and support portals that guide victims through buying and sending cryptocurrency. Bitcoin remains dominant because it is the easiest for victims to obtain, even though it is the most traceable choice for the attacker.

3.2 Laundering

Once paid, proceeds must be separated from their criminal origin. The public literature documents a consistent toolkit: moving funds through many intermediate addresses; “peel chains”, in which small amounts are repeatedly split off from a larger sum; mixing and tumbling services; cross-chain swaps that convert Bitcoin to other assets and back; and routing through privacy coins [6, 7, 10]. The common aim is to break the deterministic links that clustering exploits.

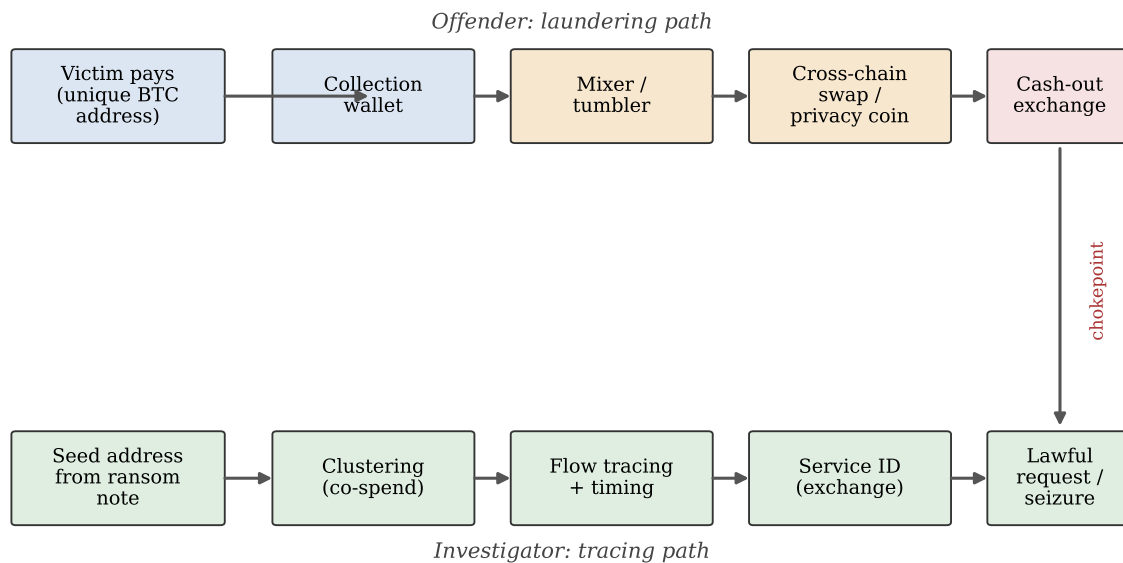


Figure 1: The offender’s laundering path (top) and the investigator’s tracing path (bottom) mirror each other. Both converge on the cash-out exchange, the ecosystem’s chokepoint, where on-chain pseudonymity most often meets a real identity through know-your-customer records.

3.3 Cash-out

At some point value must become spendable, which usually means converting to fiat currency. This is the ecosystem’s chokepoint. Conversion typically happens at exchanges, and exchanges with weak know-your-customer (KYC) controls or in permissive jurisdictions are preferred [6]. Because regulated exchanges perform identity checks, the cash-out step is where on-chain pseudonymity most often meets a real identity, which is exactly why investigators concentrate there.

4 Technical Tracing Methods

This section summarizes established public methods, at the level found in the peer-reviewed literature.

4.1 Address clustering

Clustering groups addresses likely controlled by one entity. The strongest heuristic is co-spend: inputs jointly spent in one transaction are almost certainly controlled by the same party [5]. Change-address heuristics add further links. Clustering turns thousands of individual addresses into a smaller number of entities whose behaviour can be studied.

4.2 Flow tracing and transaction-graph analysis

Once a payment address is known, investigators follow the money forward through the transaction graph, and backward to sources, using amount and timing correlations to stay on the right path through splits and merges [4]. Visualization of the resulting graph is central to both investigation and courtroom explanation.

4.3 Service identification

Attaching clusters to known services, exchanges, mixers, gambling sites, is what converts a graph into leads. When funds reach a regulated exchange, a lawful request can, in principle, obtain the account holder's identity. Identifying that an output belongs to a particular exchange is therefore often the decisive investigative step.

4.4 Cross-chain tracing

Because offenders swap between assets, investigators correlate an outflow on one chain with an inflow on another using timing and value relationships, and use exchange records where available [10]. This is harder than single-chain tracing and is an active research area.

4.5 Privacy coins: working around, not through

For Monero and similar coins, tracing individual transactions is generally infeasible by design. The practical response is to focus on the boundaries of the privacy-coin system, the exchanges where funds enter and leave, and to combine limited on-chain signals with off-chain evidence such as exchange KYC records, communications and operational-security mistakes. The realistic goal is investigative leads and corroboration, not decryption of the private ledger.

5 Case Study: The Colonial Pipeline Seizure

The 2021 Colonial Pipeline case is the clearest public demonstration of these methods, because the court filings and subsequent analysis are in the open.

In May 2021, the DarkSide group attacked Colonial Pipeline, which paid a ransom of about 75 bitcoin [1]. On 7 June 2021, the United States Department of Justice announced it had seized 63.7 bitcoin, then worth about USD 2.3 million, by following the payment on the public ledger and obtaining the private key to the address that held the funds [1]. The seizure warrant was authorized in the Northern District of California.

Two features matter for this paper. First, the seizure was possible because the Bitcoin ledger is public: investigators tracked the transfers openly and identified the address holding the proceeds [1]. Second, analysis by the firm Elliptic concluded that the seized 63.7 bitcoin corresponded to the affiliate’s share of the payment, roughly 87 percent, while the remaining share associated with the DarkSide developers was not recovered [2]. Figure 2 shows this split. It is a direct, public confirmation of the RaaS revenue-sharing model described earlier: the money divided on-chain exactly as the affiliate model predicts.

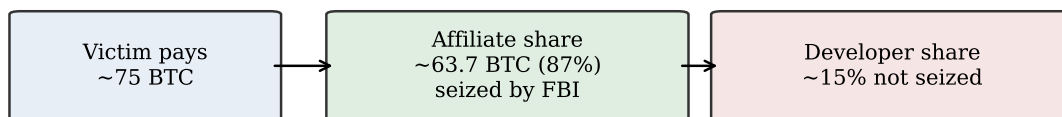


Figure 2: Approximate on-chain split of the Colonial Pipeline ransom, based on the DOJ seizure announcement [1] and Elliptic’s analysis [2]. The affiliate share was recovered; the developer share was not.

The case also illustrates the limits of tracing. Recovering the affiliate’s share required possession of a private key, and how the key was obtained was not disclosed; the developer’s share, moved differently, was not recovered. Tracing showed where the money went, but seizure and attribution needed more than the blockchain.

6 Multi-Source Attribution

The central lesson of the public record is that on-chain analysis identifies addresses and flows, but rarely proves who controls them to a criminal standard. Attribution strong enough for prosecution comes from combining several independent kinds of evidence, an approach visible across successful cases and syntheses in the literature:

- **Blockchain evidence:** transaction patterns, clustering and service interactions.
- **Communication evidence:** ransom notes, negotiation transcripts, payment instructions and the addresses embedded in them.
- **Infrastructure evidence:** IP addresses, hosting, domain registrations and server configuration.
- **Financial and lifestyle evidence:** exchange records obtained through legal process, and spending that links a cluster to a person.

Figure 3 shows how these strands converge. Each alone is usually weak; together they can meet the burden of proof.

Table 2: The four evidence types in multi-source attribution, with what each contributes and its typical weakness on its own.

Evidence type	What it contributes	Typical weakness on its own
Blockchain	Links addresses and follows the flow of funds	Rarely names the person controlling an address
Communication	Ransom notes, negotiations, and embedded payment addresses	May be pseudonymous or shared across several actors
Infrastructure	IP addresses, hosting, domains, and server configuration	Often hidden behind anonymizing services
Financial and lifestyle	Exchange KYC records and spending patterns	Needs lawful access, often across borders

The practical implication is that blockchain forensics is one input to an investigation, not a substitute for one, and that access to exchange records through lawful cooperation is frequently the decisive factor.

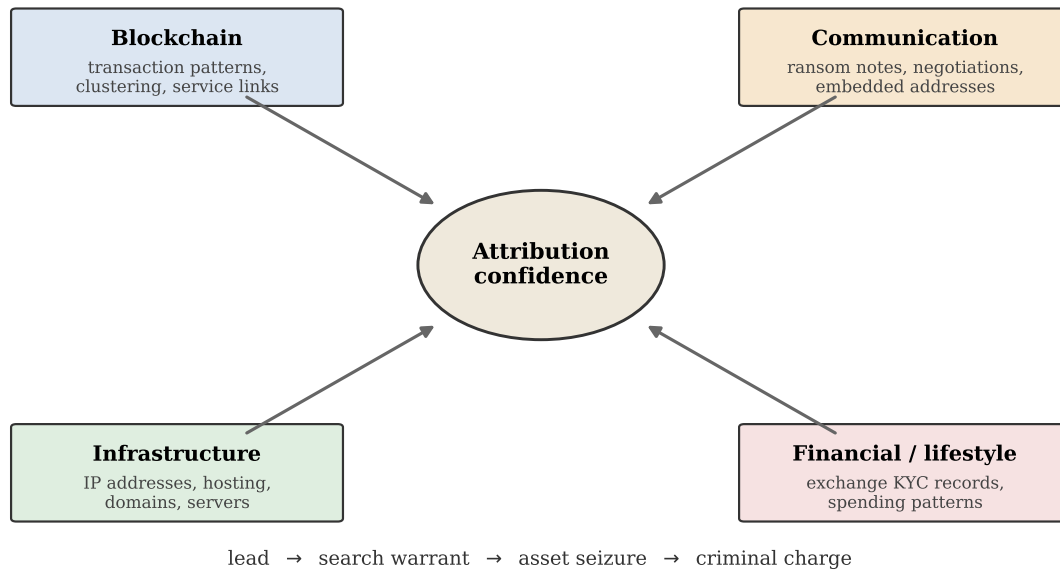


Figure 3: Multi-source attribution: blockchain, communication, infrastructure and financial evidence combine to raise confidence from an investigative lead toward a prosecutable case. Redrawn conceptually from the multi-source approach common in the literature.

7 Criminal-Justice Challenges

7.1 Jurisdiction

Ransomware operations are deliberately transnational: victim, infrastructure, exchange and offender often sit in different countries [14]. Offenders locate themselves where cooperation is weakest. Formal mutual legal assistance is slow relative to the speed at which cryptocurrency moves, which has pushed investigators toward expedited cooperation channels, joint investigation teams and parallel investigations in multiple countries.

7.2 Evidence and the courtroom

Presenting blockchain evidence in court raises real difficulties: maintaining chain of custody for data that lives on a distributed ledger, explaining technical concepts to judges and juries, qualifying expert witnesses, and meeting challenges to the reliability of tracing methods. Successful prosecutions have relied on clear documentation of method, simple visual explanations, and corroboration across the evidence types above.

7.3 Capacity

Many agencies lack the specialized tools and trained investigators that crypto cases require [15]. Responses include dedicated units, tiered models in which specialist teams support generalists, formal partnerships with analytics firms, and standardized procedures so that non-specialists can take correct first steps.

7.4 Asset recovery

Recovering funds is a race against laundering. It depends on acting before funds pass through mixers or into privacy coins, on legal mechanisms to compel exchanges to freeze assets, and on the practical problem of gaining control of private keys. The Colonial Pipeline recovery succeeded partly because the victim reported quickly, which the DOJ explicitly emphasized [1].

8 The Ransomware Economy: What the Numbers Show

Aggregate measurement of ransomware is imperfect, because it rests on attributing on-chain flows to known clusters, and totals are revised as attribution improves. Treating the figures as indicative, the published trend from Chainalysis is nonetheless striking (Figure 4). Estimated on-chain ransomware payments rose sharply in 2020 and 2021, fell in 2022, rose again in 2023, and then declined: the 2024 estimate was revised to about USD 892 million, and 2025 came in around USD 820 million, roughly 8 percent lower, even as the number of claimed attacks rose by about half [17].

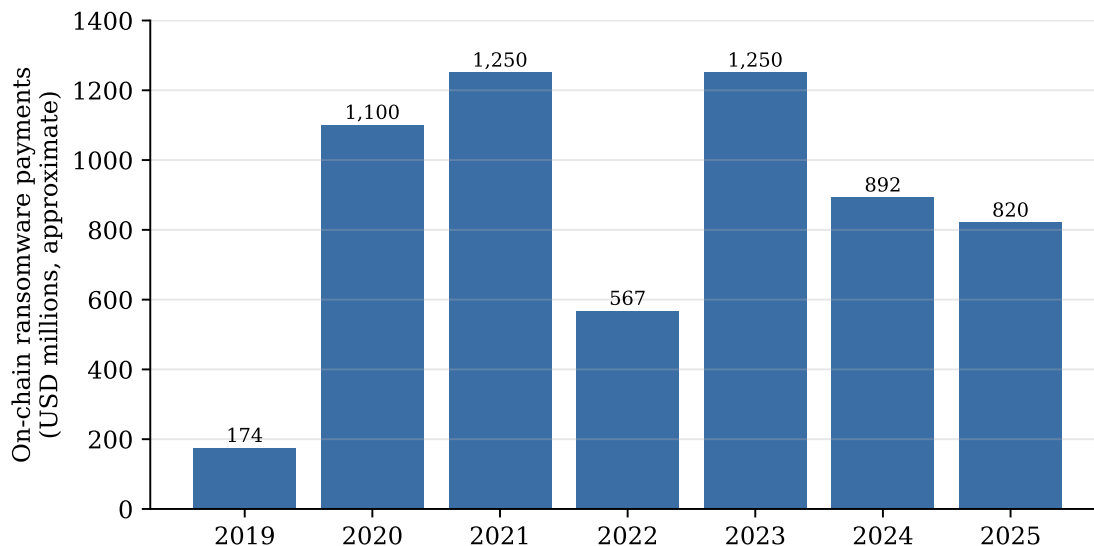


Figure 4: Approximate annual on-chain ransomware payments, from Chainalysis crypto-crime reporting [17, 18]. Figures are attribution-based estimates and are revised upward over time; they should be read as indicative, not exact.

Two published details are especially telling. The share of victims who paid fell to about 28 percent in 2025, the lowest on record [17]. Yet the median on-chain payment rose sharply, which the analysts attribute to demands anchored in fiat terms rather than to cryptocurrency price movements [17]. The combination, more attacks, fewer payers, larger individual payments, suggests an ecosystem under pressure from improved defences, greater reluctance to pay and law-enforcement action, even as the remaining successful attacks extract more per victim.

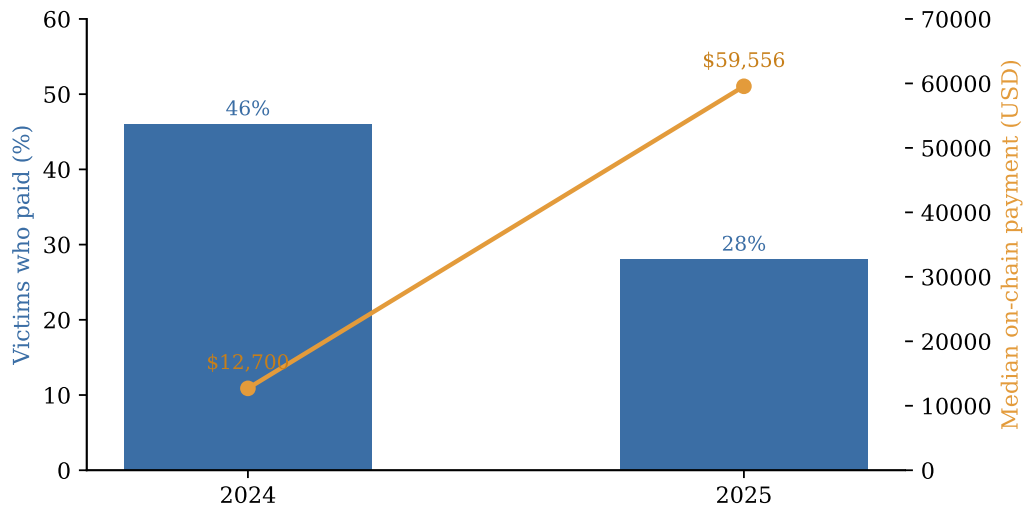


Figure 5: Two opposing signals in 2024–2025: the share of victims who paid fell to a record low near 28%, while the median on-chain payment rose sharply. Figures from Chainalysis [17]; the median rise is attributed to fiat-denominated demands rather than cryptocurrency price movements.

Reported takedowns such as Europol’s Operation Endgame are part of that pressure [17].

For this paper’s purposes the lesson is not any single number, which will be revised, but the direction: transparency plus enforcement is measurably raising the cost of the ransomware business.

9 Regional Perspective: Saudi Arabia and the GCC

The public literature on ransomware and cryptocurrency forensics is written almost entirely from United States and European viewpoints. This section adds a Gulf perspective. It is necessarily briefer than the rest of the paper, because public case data specific to the region is limited, and it draws only on published regulatory sources.

9.1 Why the region is exposed

The Gulf states are wealthy, heavily digitized and dependent on high-value critical infrastructure, which makes them attractive targets for financially motivated and state-aligned attackers alike. Saudi Arabia in particular has been the target of major destructive attacks on industrial organizations. Rapid digitization under Vision 2030 widens the attack surface for ransomware even as it improves services.

9.2 Regulatory context

Several features of the regional framework bear on ransomware response. In Saudi Arabia, computer-related offences including extortion and unauthorized access are addressed by the Anti-Cyber Crime Law of 2007 [19]. Financial institutions operate under the Saudi Central Bank (SAMA) Cyber Security Framework, which since 2022 has included threat-management and cyber-threat-intelligence requirements that push regulated firms to detect and report incidents [20, 21]. Critical infrastructure and government entities fall under the National Cybersecurity Authority’s controls. In the UAE, the cybersecurity authorities have likewise built incident-response and threat-intelligence capacity. These frameworks matter for ransomware because early reporting and good telemetry are exactly what made recoveries like Colonial Pipeline possible.

9.3 Cryptocurrency and cash-out in the region

A practical point for regional investigators is that cash-out is the chokepoint everywhere, including the Gulf. Regional virtual-asset regulation, such as the frameworks emerging in the UAE, affects where offenders can convert proceeds and where lawful requests for exchange records can succeed. Harmonized virtual-asset regulation and KYC enforcement across the GCC would reduce the room for regional cash-out.

9.4 What would help

Three regional priorities follow from the technical picture. First, build blockchain-forensic capacity in national cybercrime units, so that investigators can act at the speed cryptocurrency demands. Second, strengthen formal cooperation channels with exchanges and with counterpart agencies, since attribution and recovery depend on cross-border access to records. Third, use the reporting culture that SAMA and NCA frameworks already require, so that victims report quickly and preserve the evidence that makes tracing possible.

10 Recommendations

10.1 For law-enforcement agencies

1. Build tiered blockchain-forensic capability, from basic awareness for all investigators to specialist units, and adopt standardized first-response procedures for crypto evidence.
2. Establish working relationships with exchanges and analytics firms before incidents occur, to speed lawful requests and asset freezes.
3. Treat attribution as a multi-source problem, combining on-chain analysis with communication, infrastructure and financial evidence.
4. Prioritize speed in asset recovery, acting before funds reach mixers or privacy coins.

10.2 For prosecutors and courts

1. Develop clear standards for admitting and explaining blockchain evidence, including expert-witness qualification and simple visual presentation.
2. Invest in judicial education on how cryptocurrency tracing works and what it can and cannot prove.

10.3 For policymakers and regulators

1. Strengthen and harmonize KYC and anti-money-laundering requirements for exchanges, since cash-out is the ecosystem's chokepoint.
2. Support expedited cross-border cooperation for time-critical crypto investigations.
3. Encourage or require prompt incident reporting, which materially improves the odds of tracing and recovery.

10.4 For the cryptocurrency industry

1. Maintain effective monitoring for ransomware-linked flows and respond quickly to lawful requests and freeze orders.
2. Cooperate through formal public-private channels rather than case by case.

10.5 For potential victims

1. Maintain tested, offline backups so that paying is rarely the only option.
2. Report incidents to law enforcement early, and preserve evidence, since early reporting was decisive in documented recoveries [1].

11 Limitations

Review, not original casework. This paper analyses no confidential cases and produces no new blockchain measurements. Its contribution is synthesis and regional framing.

Attribution-based statistics. Aggregate figures come from analytics firms, rest on proprietary attribution, are lower bounds and are revised over time. They are used as indicative evidence of trends, not as exact measurements.

Limited regional case data. Public, verifiable ransomware case data specific to Saudi Arabia and the GCC is scarce, so the regional section relies on regulatory sources and general exposure rather than on documented local cases.

Not legal advice. Statements about law rely on published legislation and commentary; the binding Arabic texts and qualified counsel govern in practice.

Fast-moving field. Both laundering techniques and tracing methods evolve quickly, so specific tactics described here will date even though the structure of the problem is stable.

12 Conclusion

Cryptocurrency gave ransomware a global, bank-independent payment channel, but the same public ledgers that enable payment also enable investigation. The documented record, above all the Colonial Pipeline seizure, shows both sides clearly: transparent blockchains let investigators trace and sometimes seize funds, the RaaS affiliate model is visible in how payments split on-chain, and yet attribution and recovery still depend on private keys, exchange records and cross-border cooperation that the blockchain alone does not provide. The published trend is encouraging: more attacks but fewer payers and falling aggregate payments suggest that better defences, greater reluctance to pay and sustained enforcement are working. For Saudi Arabia and the Gulf, the priorities are practical: build blockchain-forensic capacity, use the reporting culture the regional frameworks already require, and press for harmonized virtual-asset regulation so that the cash-out chokepoint stays narrow. Progress against ransomware will keep coming from joining technical tracing to criminal-justice cooperation, not from either alone.

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